

**ATTACHMENT 34 – MEMORANDUM FROM GANNETT FLEMING, INC.
DATED OCTOBER 12, 1998 REGARDING THE USE OF FACTOR OF SAFETY
OF TWO (2) APPLIED TO MANUFACTURER’S ULTIMATE CAPACITIES
(2 pages)**

**CENTRAL ARTERY (I-93)/THIRD HARBOR TUNNEL (I-90) PROJECT
FORT POINT CHANNEL CROSSING - D09B2**

MEMORANDUM

To: P. J. Falk

From: E. V. DiSalvio [REDACTED]

Date: 10/12/1998

Re: Meeting with B/PB on 10/9/98

Attendees: P. Das - B/PB B. Richards - B/PB
J. Bartoletti - B/PB P. J. Falk - GF
E. V. DiSalvio - GF
[REDACTED]

This meeting focused on the use of adhesive anchors which will be used to attach the hanging ceiling supports to the concrete tunnel roof.

P. Das directed us to incorporate the following design criteria into our present design.

1. For the one-hanger out scenario:
 - a. use a factor of safety of two (2) applied to manufacturer's ultimate capacities to arrive at allowable capacity for adhesive design.
 - b. use 0.85 Fy on the threaded bolt material design. This requires determining a proportional allowable stress increase using the AISC formulas which are based on using Fu to determine allowable stresses for threaded bolt materials. Do not exceed 1.00 Fy for allowable stress.
2. Stainless Steel will no longer be used for threaded fasteners. Use A36 material.
3. B/PB may limit embedment depth to 5" or 5-1/2". Pete, please verify.
4. P. Das will review research papers provided by GF pertaining to cracked concrete and notify GF on Monday October 12, 1998 as to his decision if we should consider the cracked concrete in the design of the adhesive anchors.

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